Network Access and Market Access

ECON 895 Spatial Techniques in Empirical Economics

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Plan for tonight

Least-cost distance

Dijkstra's algorithm

The cost surface

Where market access comes from

Building it with Claude

Jewish communities and city growth

The exclusion radius and a rivers counterfactual

Your paper

We break around 8:45. Claude Code does the computing tonight. Have it installed and the course project pulled.

Exercise 1

- Exercise 1 was due by email at 7:20 tonight. Send it now if yours is not in.
- I return it with comments before Week 8. We discuss the audit half once every submission is graded.
- Exercise 2 goes out next week with the climate material and is due in Week 9.

Least-cost distance

Week 2 measured distance as a straight line

- Dittmar's instrument was the straight-line distance from Mainz.
- A merchant in 1500 moved on roads, down rivers, and by sea. Each mode cost a different amount per kilometer.
- Tonight distance is the cost of the cheapest route. Market access adds up the population of every other city, each divided by that cost.

Dijkstra's algorithm

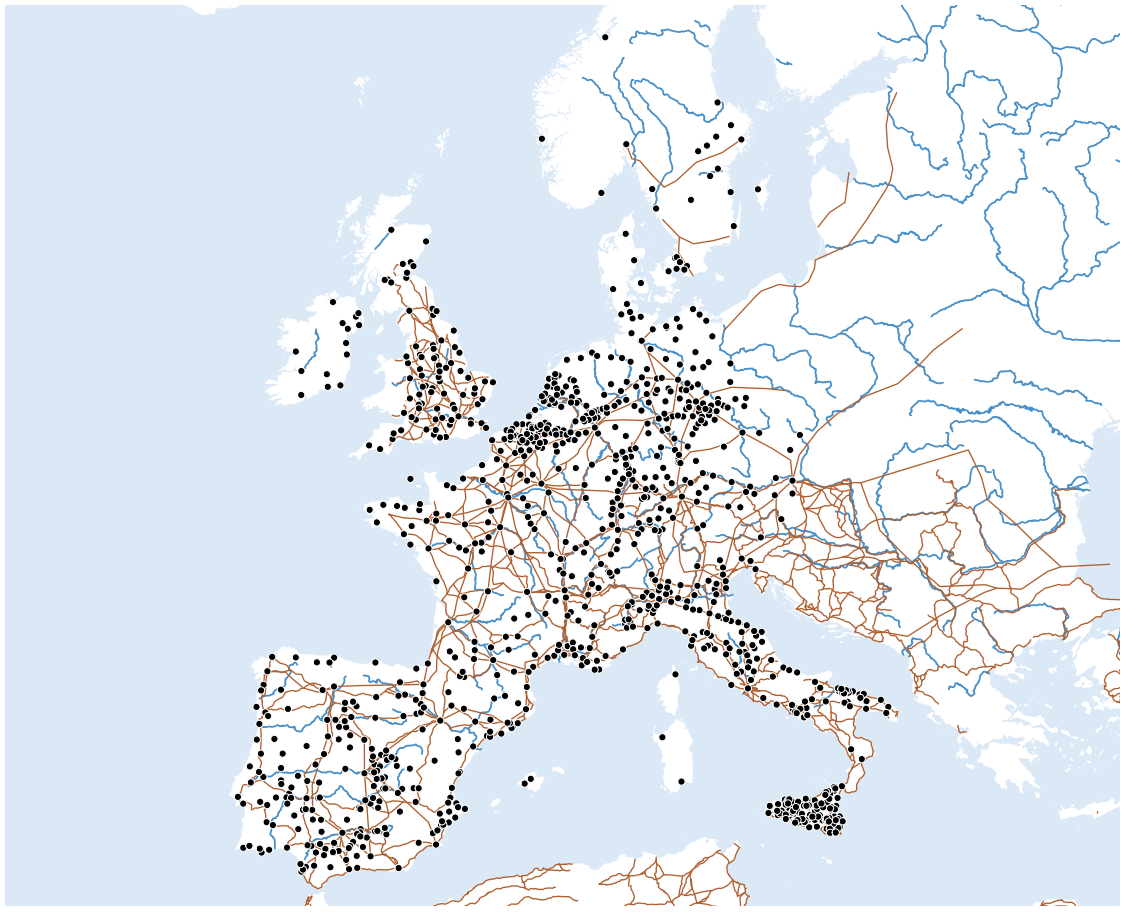
Stepping through Dijkstra's algorithm

noeldjohnson.github.io/teaching/dijkstra

- Part 1 steps the algorithm through a six-node network, one settled node at a time, with the table of tentative costs.
- Part 2 routes across a raster in which each cell is a node joined to its eight neighbors. This is the graph that `gdistance` builds.

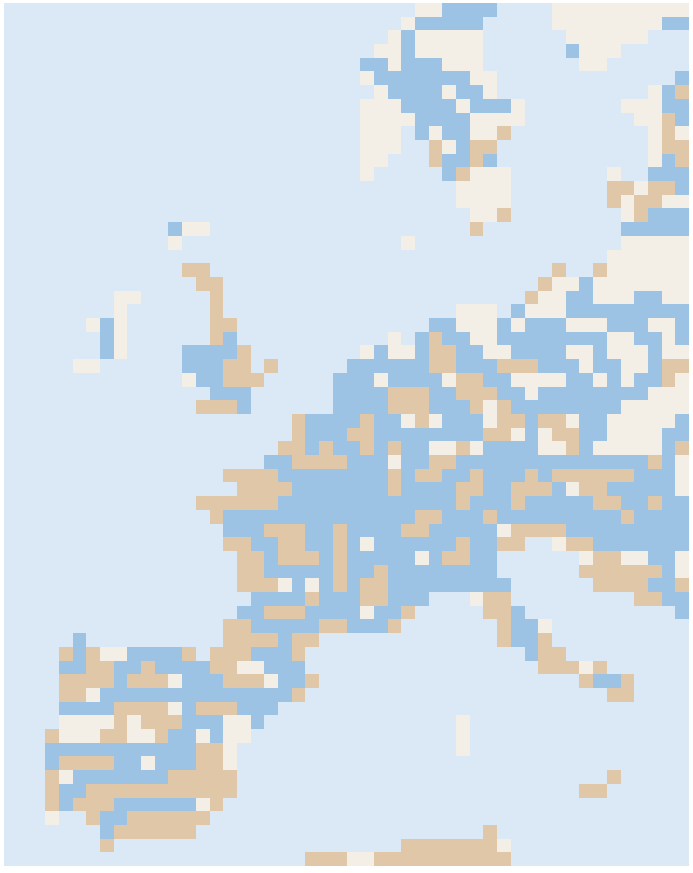
The cost surface

Four layers



Roman roads and medieval trade routes in brown, rivers and seas in blue, and the 1,017 cities with population data as black dots. All four layers ship in one equidistant conic CRS with meters as units.

Each cell takes the cheapest mode that crosses it

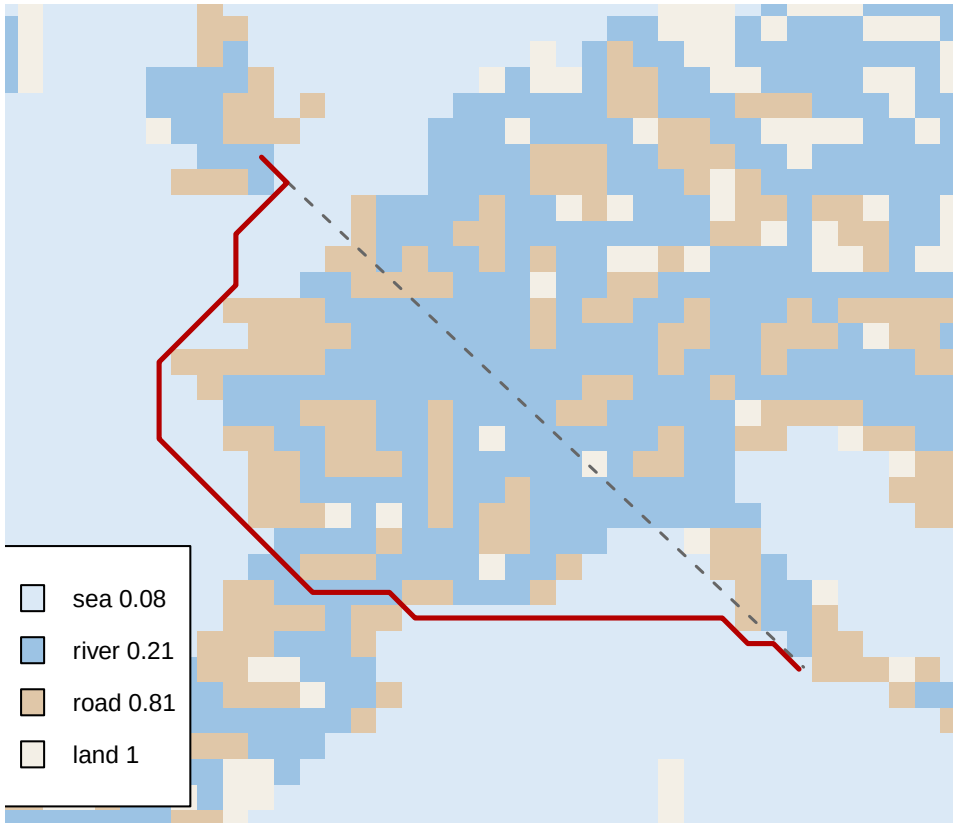


Relative cost of carrying a ton one kilometer

overland	1
road or route	0.81
river	0.21
sea	0.08

The grid has 50 km cells. A route's cost adds up, cell by cell, the distance traveled times that cell's cost. Every distance is therefore in kilometers of overland carrying.

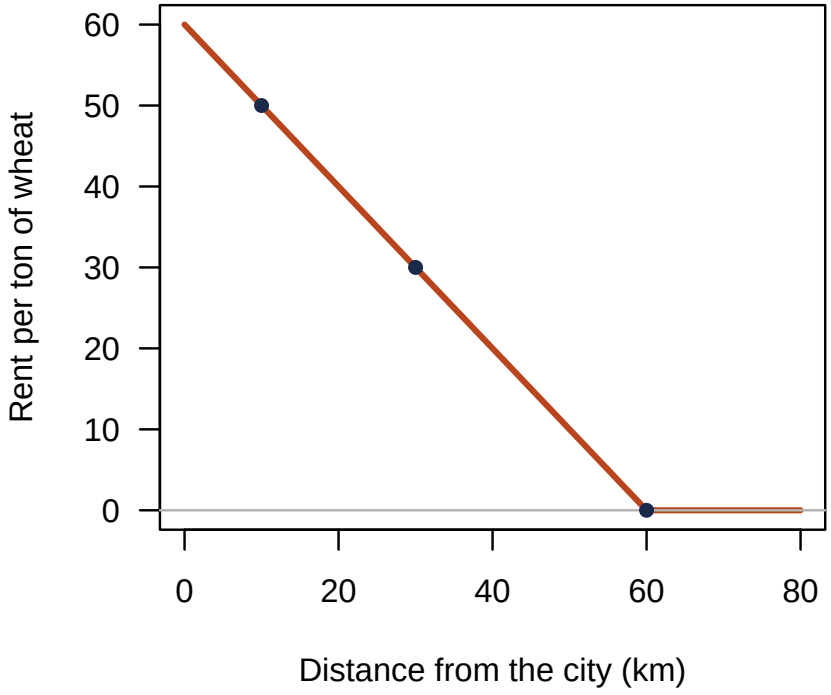
London to Rome



The dashed line is 1,431 km. The least-cost path is 2,111 km long and costs 211 km-equivalents. Going along the dashed line instead costs 405 km-equivalents, which is 1.9 times as much. The cheapest route leaves the Channel, rounds Brittany, enters the Garonne at Bordeaux, crosses to the Mediterranean, and follows the coast to Rome.

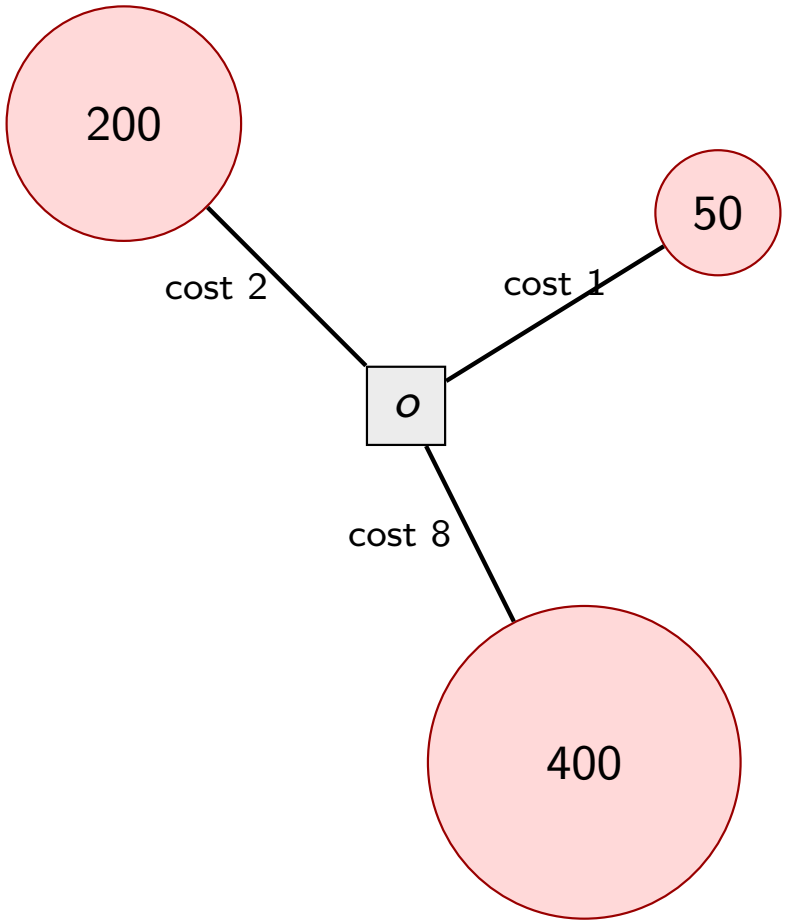
Where market access comes from

Land near a market is worth more



- Wheat sells for 100 a ton in the city. Growing a ton costs 40. Carting it costs 1 a ton for every kilometer.
- A farm 10 km out keeps 50 a ton, one 30 km out keeps 30, and one 60 km out keeps nothing.
- What a farmer keeps is what the land is worth to rent. Fogel argued that American land values reflect this cost of getting crops to market.

Market access weights each market by its size and its cost to reach



A place o can sell to three cities, labeled by population. Market access adds them up, each divided by the cost of reaching it.

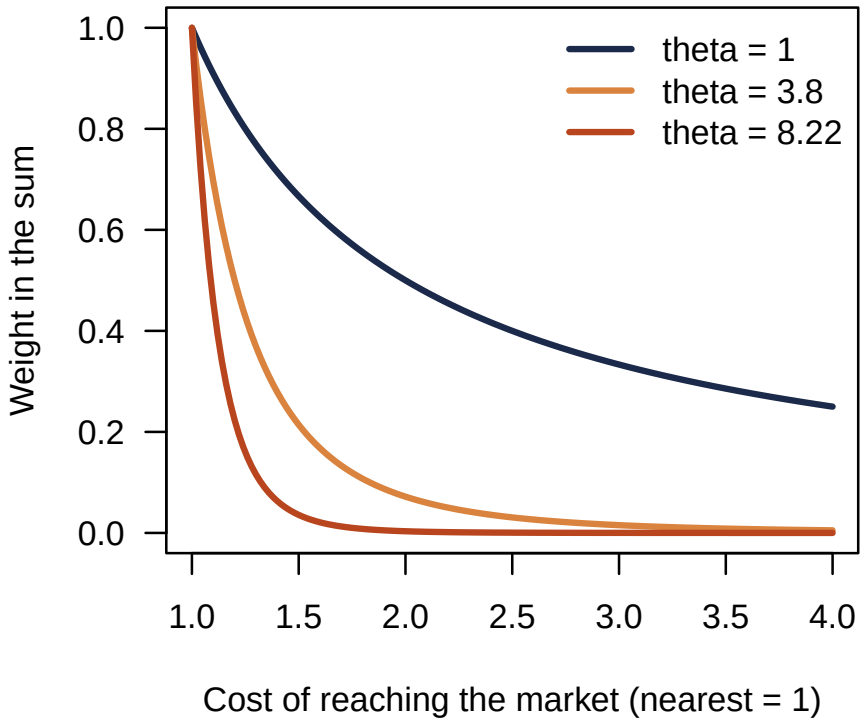
$$MA_o = \sum_d \frac{N_d}{\tau_{od}^{\theta}}$$

With $\theta = 1$,

$$\frac{200}{2} + \frac{50}{1} + \frac{400}{8} = 100 + 50 + 50 = 200.$$

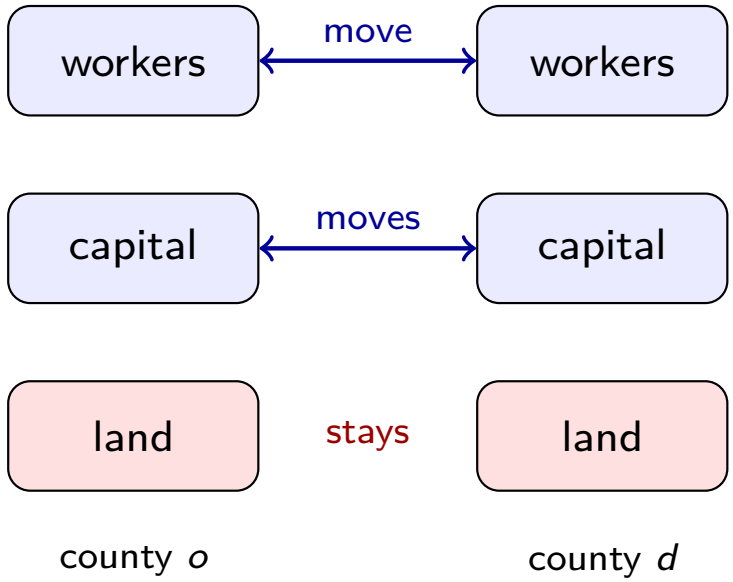
The largest city and the smallest each add 50 because the largest is eight times as populous and eight times as costly to reach.

Theta sets how fast a market's weight falls with cost



- A market twice as costly to reach as the nearest one gets weight $2^{-\theta}$.
- That is 0.5 at $\theta = 1$, 0.07 at $\theta = 3.8$, and 0.003 at $\theta = 8.22$.
- θ is the trade elasticity. A 1 percent rise in shipping cost lowers trade by θ percent.
- Johnson and Koyama use 1, Donaldson's India paper estimates 3.8, and Donaldson and Hornbeck (2016) estimate 8.22.

Workers and capital move and land rent absorbs market access



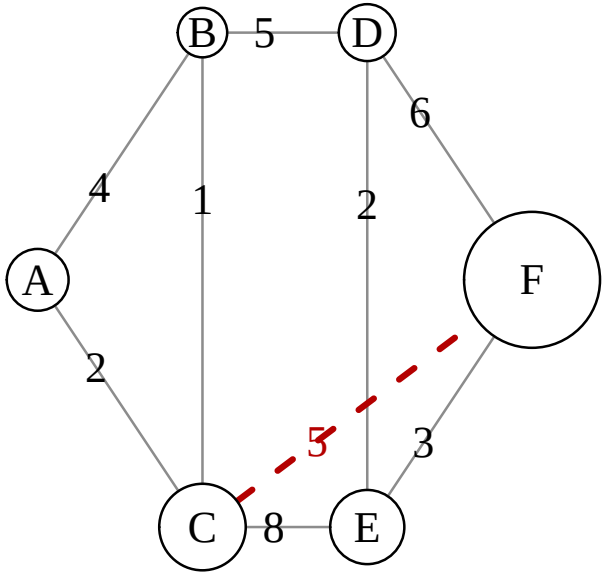
- Workers move until they are equally well off everywhere. A county with cheap goods can pay a lower wage and still keep them.
- Capital moves until it earns the same return everywhere.
- Land cannot move. Whatever a county's access to markets is worth ends up in its land rent.

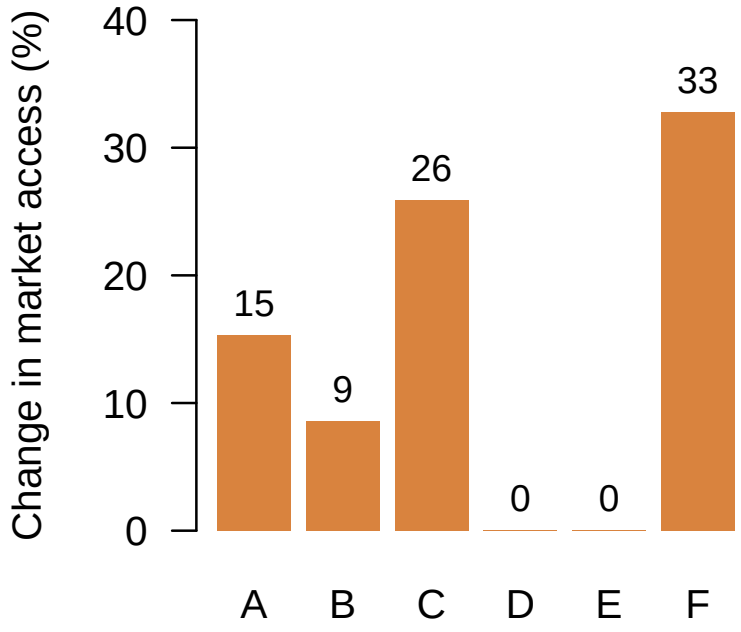
Solving the model gives Donaldson and Hornbeck's equation (10).

$$\ln(\text{land rent}_o) \;=\; \text{constant} \;+\; \beta \ln MA_o \;+\; \text{productivity}$$

β is how much land rent rises when market access rises. With their farm shares of land and labor the model predicts 0.62. A 10 percent rise in market access then raises rent by about 6 percent. Their regression estimates 0.51.

A new link changes market access beyond its two ends

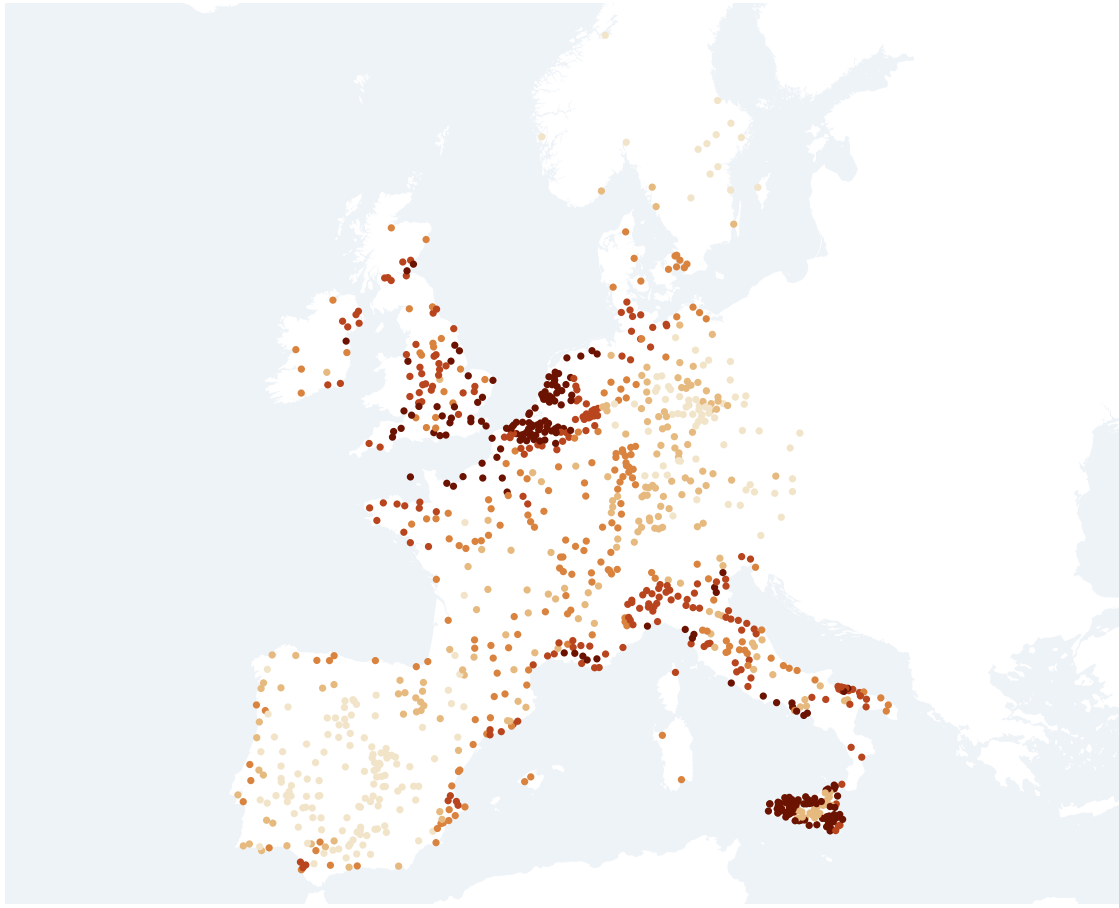




The six-node network from the Dijkstra page, with populations A 10, B 5, C 20, D 8, E 15, and F 40, drawn by size. The new link from C to F at cost 5 is dashed. Market access rises at C and F and also at A and B, whose cheapest routes to F now run through C. D and E are unchanged. A railroad changes market access in counties it never reaches. Donaldson and Hornbeck therefore regress land values on market access in place of a railroad indicator.

Try it at noeldjohnson.github.io/teaching/dijkstra, Part 3.

The map of market access in 1800



Cities colored by quintile of market access in 1800, darkest highest. Coasts and the Rhine corridor are dark. Inland Iberia and the Alps are light.

Building it with Claude

Starting the project in Claude Code

Copy `examples/market_access_project/` to `~/ma-demo`, open a terminal there, and start `claude`. The folder holds the data and a `CLAUDE.md` that specifies in words every choice made in the cost-surface and market-access sections. It has no scripts.

The request

```
Read CLAUDE.md. Write scripts/01_cost_surface.R and scripts/02_market_access.R to
build the cost surface and least-cost matrix it describes and to compute market
access for 1800. Run both. Report the grid size, the number of cells of each cost,
the number of city pairs that share a cell, the least-cost distance from LONDON to
ROMA, and market access in 1800 for AMSTERDAM, LONDON, PARIS, MADRID, and WIEN.
```

Read the script it writes before reading any number it reports.

What the answer has to show

These numbers were computed with the same choices when the slides were built. Claude's report should match them.

grid	63 by 50 cells
sea cells	1,953
river cells	565
road cells	340
overland cells	292
city pairs sharing a cell	889
London to Rome	211 km-eq.

Market access in 1800	
Amsterdam	274
London	186
Paris	162
Madrid	88
Vienna	86

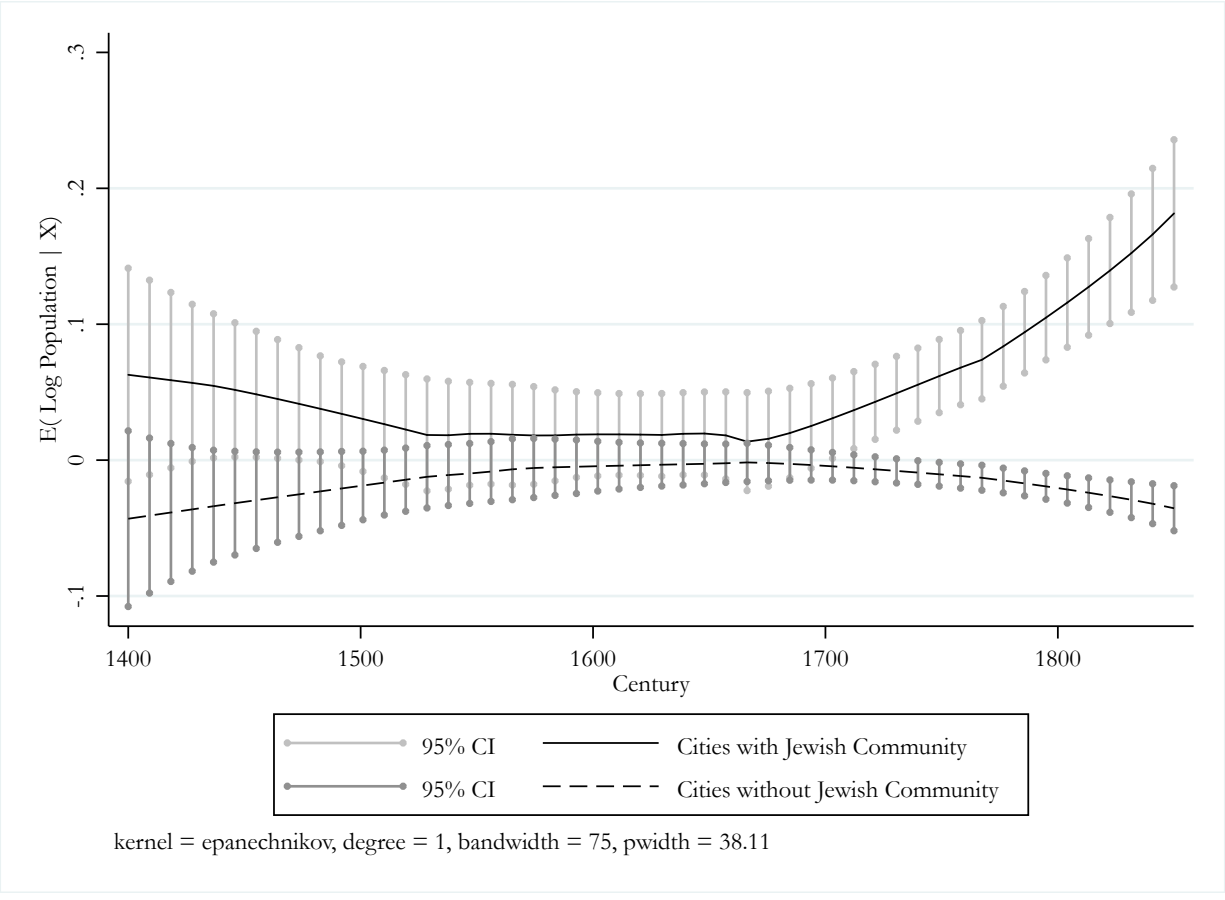
If a number differs find the line in the script that makes the difference. The 889 pairs sharing a cell are where a model is most likely to depart from the specification. At $\tau = 0$ the weight $1/\tau$ is infinite.

Break

Ten minutes.

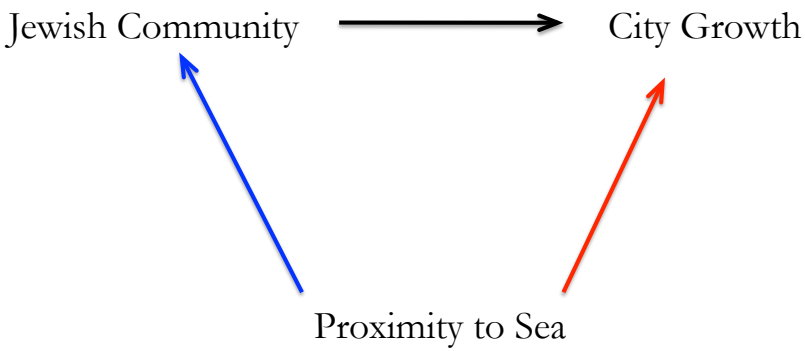
Jewish communities and city growth

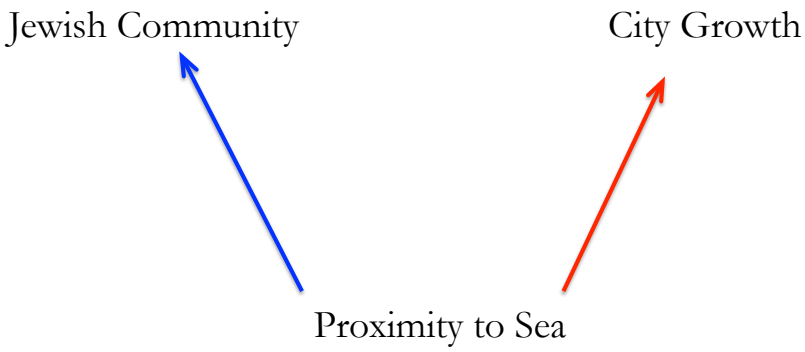
Cities with Jewish communities pulled ahead after 1700



Log city population by year for cities with and without a Jewish community, 1400–1850, net of city and year fixed effects and controls, with 95 percent confidence intervals. Extended sample. From Johnson and Koyama (2017).

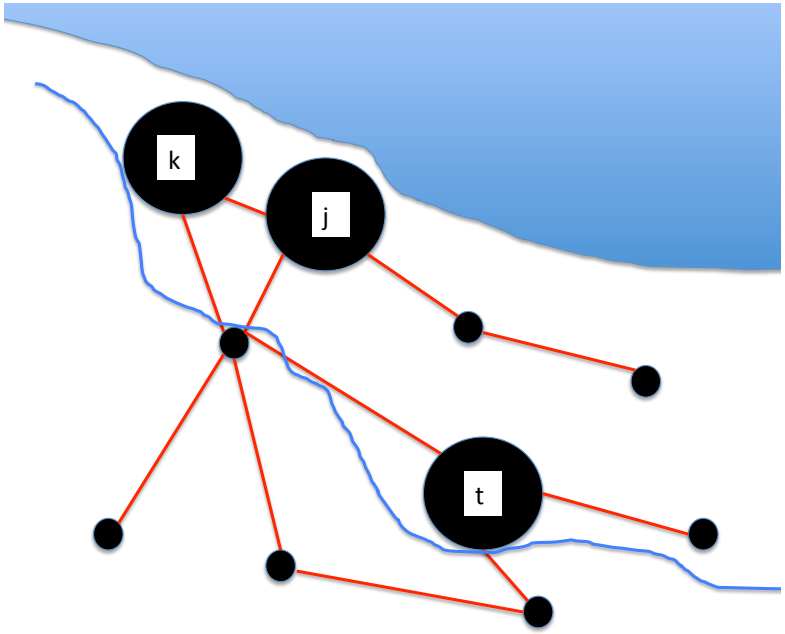
A city's own Jewish community is not randomly assigned





- A coastal city was more likely to host a Jewish community and more likely to grow. The bottom diagram has no arrow from community to growth. The coast alone produces the correlation there.
- A strong merchant class could keep Jews out, as in Turin until 1424 and Florence until 1437. Expulsions followed downturns. If Jews were invited to declining cities, OLS is biased down. If they chose prosperous ones, it is biased up.
- Controls handle the factors we can measure. The instrument is for the ones we cannot.

Network access counts the communities a city can reach cheaply



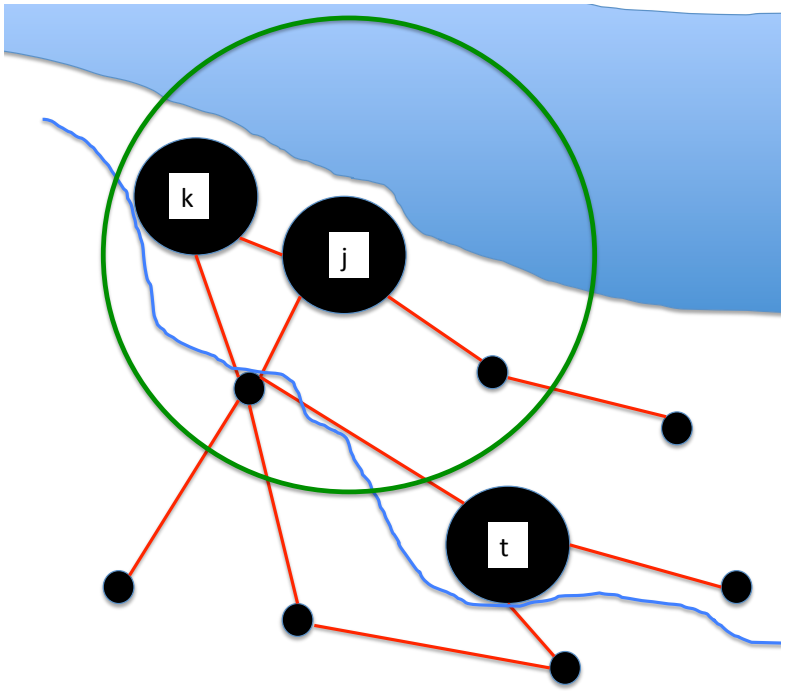
$$NA_{jt} = \sum_{i \neq j} JC_{it}\, \tau_{ji}^{-\sigma}$$

- JC_{it} is 1 if city i has a Jewish community in period t. τ_{ji} is the cost of the cheapest route from j to i. The paper sets $\sigma = 1$.

- Moving a ton one kilometer cost 8.8 kg of grain by porter, 3.9 by cart, 0.99 by river, and 0.3–0.4 by sea (Bairoch 1990). The paper divides each by the porter cost.

- City j reaches k and t along roads and rivers. City k shares j's coast and river. Its community and j's growth can therefore share a cause.

The Jewish network IV drops every community within D kilometers



$$\text{IVNetwork}_{jt} = NA_{jt} \Big|_{d_{ji} > D}$$

- The green circle has radius D around j. Community k inside it is left out. Community t outside it still counts, weighted by least-cost τ.
- d_{ji} is straight-line distance. The instrument predicts a community in j from the network more than D km away.
- The assumption is that whatever drives growth and Jewish presence in j is unrelated to Jewish presence more than D km away.
- The paper builds it for $D = 0$, 50, 100, 250, and 500 km.

Johnson and Koyama's estimates across the radius

	Dependent Variable: Log City Population				
	Panel A: Jewish Network IV				
	(1)	(2)	(3)	(4)	(5)
	All Cities	*>50km*	*>100km*	*>250km*	*>500km*
Jewish Community	0.467***	0.467***	0.491***	0.522***	0.348
	(0.0615)	(0.0643)	(0.0695)	(0.100)	(0.367)
Year FE	Yes	Yes	Yes	Yes	Yes
Controls X Year FE	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes
N	2889	2889	2889	2889	2889
First Stage F-stat	677.05	676.60	540.73	179.84	11.84
	Panel B: Expulsion IV				
	(1)	(2)	(3)	(4)	(5)
	All Cities	*>50km*	*>100km*	*>250km*	*>500km*
Jewish Community	0.125	0.381**	0.864***	2.777***	-0.755
	(0.165)	(0.172)	(0.255)	(0.918)	(0.503)
Year FE	Yes	Yes	Yes	Yes	Yes
Controls X Year FE	Yes	Yes	Yes	Yes	Yes
City FE	Yes	Yes	Yes	Yes	Yes
N	2889	2889	2889	2889	2889
First Stage F-stat	51.78	48.36	28.40	9.51	11.17

Table 2 of Johnson and Koyama (2017). Main sample, 1400–1850. Columns move the exclusion radius D from 0 to 500 km.

The network IV estimate is stable until the first stage weakens

- The network IV coefficient stays between 0.47 and 0.52 from 0 to 250 km. At 500 km it is 0.35 and insignificant, with a first-stage F of 12 against 677 at $D = 0$.

- At $D = 0$ the expulsion IV estimate is 0.125 and insignificant, with a first-stage F of 52. Many expulsions were national. Nearby cities fell under the same decree. The estimate rises with D and turns negative and insignificant at 500 km.

- Adding market access as a control lowers the 100 km network IV from 0.491 to 0.242 (their Table 3).

The exclusion radius and a rivers counterfactual

Market access beyond D kilometers in the course data

The request

```
Add scripts/03_beyond_D.R. For D = 0, 50, 100, 250, and 500 km, compute market
access in 1800 using only cities more than D km away in a straight line. Report
the correlation of its log with the log of full market access for each D.
```

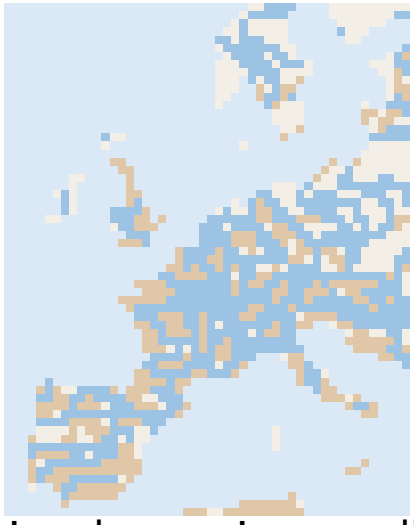
Population stands in for the Jewish community indicator, which the course data do not have. The correlation plays the role of the first stage in Table 2, which weakens as D grows.

D (km)	0	50	100	250	500
correlation	1.00	0.96	0.91	0.81	0.38

in the terminal

What the counterfactual has to satisfy

Rivers at 0.21



Rivers costed as land

Removing the river layer raises a cell's cost only where a river was its cheapest mode. A cell a river shares with the sea stays at 0.08 and one it shares with a road goes to 0.81.

1. No cost between two cities can fall. The CSV holds only market access. Check this condition in the script.
2. No city's market access can rise.
3. The largest falls should be in cities on rivers far from the sea.

Claude's report checks the second condition and the map of the result checks the third.

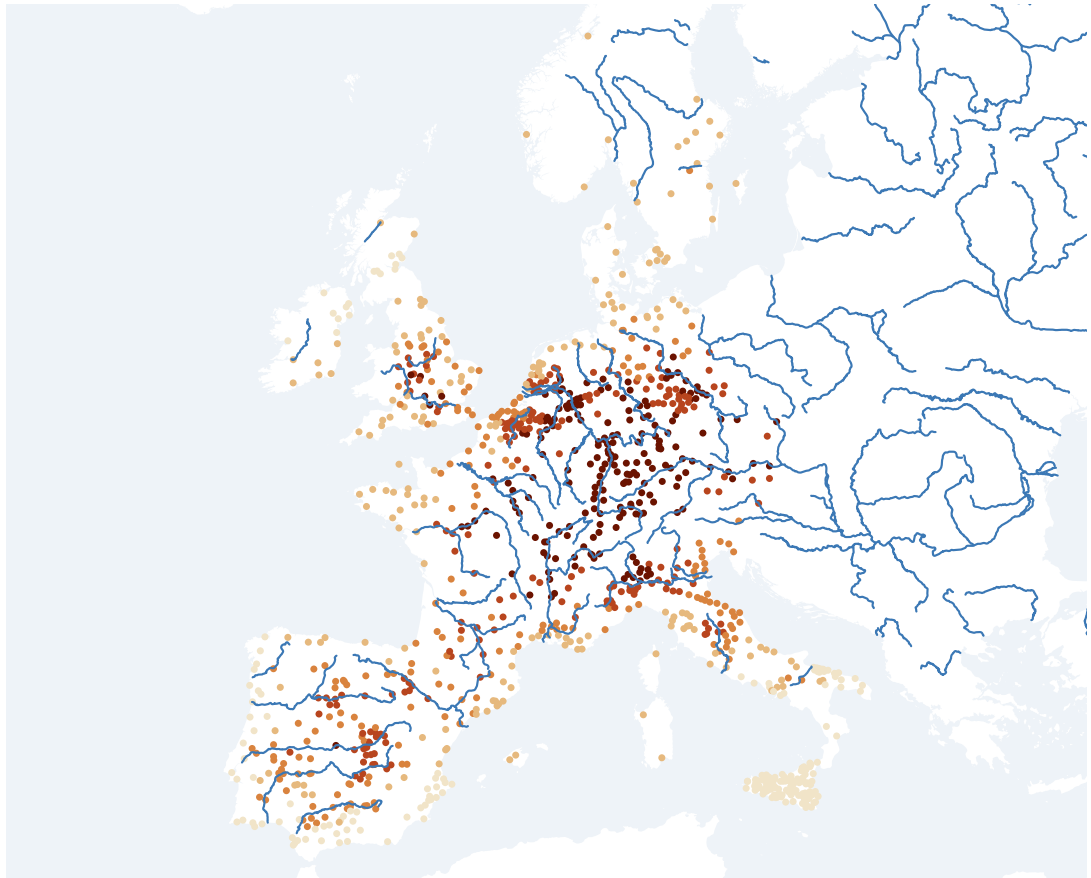
The rivers-as-land counterfactual

The request

```
Add scripts/04_rivers_as_land.R. Rebuild the cost surface with rivers costed as
land, recompute market access for 1800, and write
Data/processed/ma_rivers_as_land.csv with columns city_jjk, ma_1800, and
ma_1800_cf. Report whether any city's market access rises, the mean percent
change, and the five cities that lose most.
```

The answer should report that no city rises and that market access falls 37 percent on average. The five cities that lose most are Dijon, Belfort, Mulhouse, Besançon, and Salins-les-Bains.

Market access falls most in Germany and eastern France



Cities colored by quintile of the fall in market access when rivers cost as much as land, darkest the largest fall, with rivers in blue. German cities lose 54 percent on average, French cities 44, and Italian cities 23.

Your paper

Which network does your question run on?

- A cost surface needs a mode list with costs, a grid, and the places. Roads, rivers, rail, shipping lanes, and terrain all fit the same construction.
- The railway you traced on the 1911 Gold Coast map in Week 5 is a network layer. A cost surface for the Gold Coast needs that railway, a rail cost, and a land cost.
- Market access turns the whole network into one number per place. Every choice behind the surface changes that number. Your paper has to report each one.

Before next week

In your copy of the project ask Claude to change one of the mode costs, the cell size, or θ. Check what it changed and report how market access moved.

Week 8

- Week 8 is climate data and spatial interpolation, with the persecutions paper. Interpolation builds a surface from measurements at points.
- Exercise 2 goes out next week and is due in Week 9. Like Exercise 1 it has a build half and an audit half.
- Exercise 1 comes back with comments before then.

References

Donaldson, D. and Hornbeck, R. (2016). Railroads and American economic growth: A "market access" approach. *The Quarterly Journal of Economics*, 131(2):799–858.

Johnson, N. D. and Koyama, M. (2017). Jewish communities and city growth in preindustrial Europe. *Journal of Development Economics*, 127:339–354.